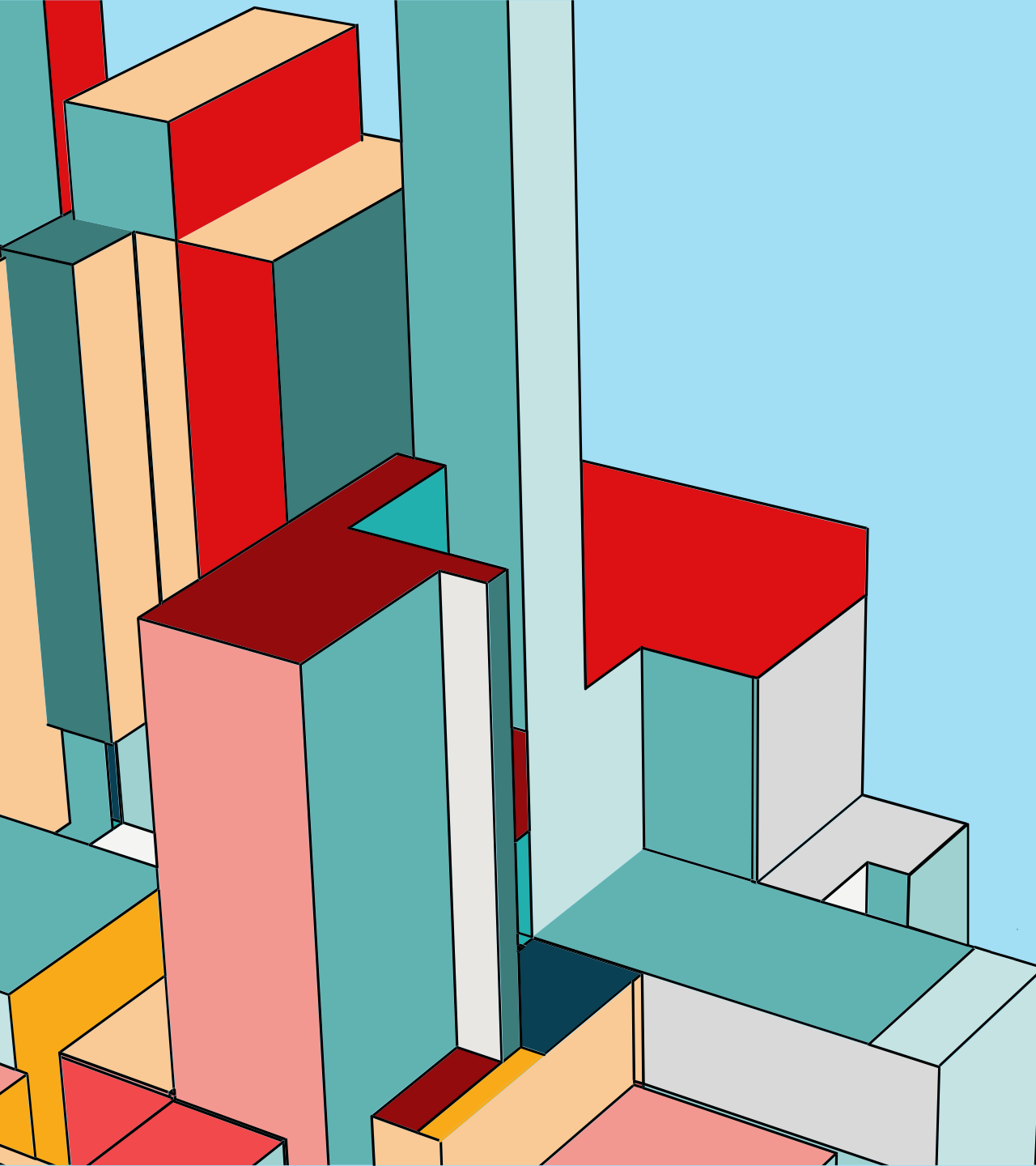




ENGR 315 FINAL PROJECT PROPOSAL: WEARABLE DEVICE- STRESS & EXERCISE INDUCED OPERATION

Austin Conley, Justin
Criscuolo, Paul Clossey

TASK #1: DATA OF INTEREST

Interest: The data was found by using physiological signals collected during structured acute stress induction and aerobic and anaerobic exercise sessions via wearable devices.

URL: [Wearable Device Dataset from Induced Stress and Structured Exercise Sessions v1.0.1](#)

Data Collection:

- The metrics collected: Blood volume pulse (BVP), skin temperature, heart rate, electrodermal activity, accelerometer data
 - Recorded with the Empatica E4 (a research-grade) wearable

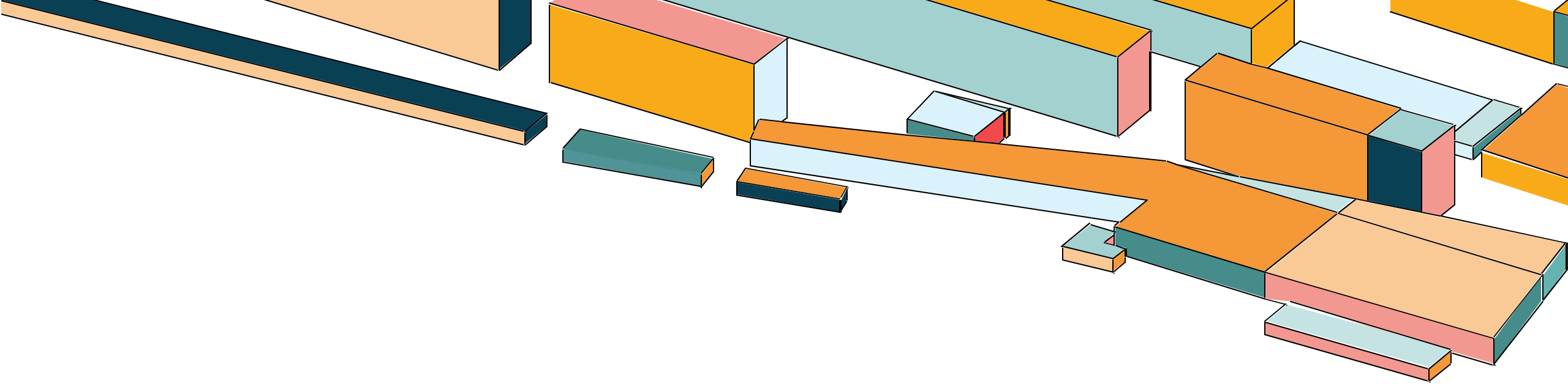
The methods to induce stress involved mathematical and emotional tasks within a set timeframe, interleaved with rest periods.

The exercise involved were distinct routines on a stationary bike, created for aerobic and anaerobic activities.

- The dataset includes records from 36 healthy volunteers for stress sessions, 30 for aerobic exercise, and 31 for anaerobic exercise

Why?

- This will allow us to see the effects gender, age, or activity level in life can dictate how their body responds to stress and different exercises. Hopefully there is some indication of healthier lifestyles and ways to exercise based off the questions we answer from the data.



TASK #2 QUESTIONS OF INTEREST

QUESTION #1

- On average, how do the heart rates of participants who reportedly do physical activity often, compare to participants who don't do physical activity often; during aerobic exercise?

This question is interesting because the data in this set might be useful in demonstrating that doing physical activity often might help your body maintain a lower heart rate during periods of prolonged exercise. This is an inferential question because it challenges the team to come to a conclusion based on sample data; the team will compare two groups using sample data to make a claim about people in general. The variables of interest might be: "Does physical activity regularly" then "Yes/No" for each participant, found in subject-info.csv; HR.csv values from the Aerobic folder for each participant; perhaps, some of the other signal information from Aerobic, such as BVP.csv, IBI.csv, and tags.csv.

QUESTION #2

- Does aerobic exercise get better as we age or get worse for individuals who do workout often?

This question is interesting because the data in the set might be useful in demonstrating that aerobic exercise requires more physical exertion (i.e. a higher heart rate) when you are older even when you work out often, or can an older person perform better than a younger person in this metric. This is an inferential question because it requires the team to compare two groups (young and older) using sample data to make a claim about people in general. The variables of interest might be: "Age", "from 18-25 (younger)" then "from 26-31 (older)"; "Does physical activity regularly" then "Yes (must be yes because we want to compare those who exercise often)", found in subject-info.csv; HR.csv, BVP.csv, IBI.csv, and tags.csv from the Aerobic folder.

QUESTION #3

- Does arguing for the opposite opinion make a person's heart rate depress or jump when arguing for the opposite opinion?

This question is interesting because the data in this set might be useful in demonstrating that arguing for something you do not agree with can increase or decrease your heart rate. Many times when people are passionate about a topic they may argue or get into an elevated discussion about it. We wonder if when a person is forced to argue against what they are passionate about does this induce higher or lower heart rates? This is an inferential question because it challenges the team to figure out a conclusion to this question based on the sample data. The variables of interest might be: "Stress inducement"; HR.csv.

QUESTION #4

- For individuals over the average weight in the U.S., how does anaerobic exercise affect heart rate compared to those under average?

This question is interesting because the data in this set might be useful in demonstrating that those who are over the average weight in the U.S. experience higher heart rates during anaerobic exercise than those who are under average. Obesity is commonly talked about in the health field and from this study we want to see if people who weigh more than the average person in the U.S. have higher heart rates. This is an inferential question because it challenges the team to derive a conclusion about the correlation between weight and heart rate during exercise. The variables of interest might be: "Weight" and compare those values to the U.S. average, if below average: put in the below group, if above average: put in the above group; HR.csv in Anaerobic folder for each participant recorded in each group.

QUESTION #5

- Do anaerobic exercise or aerobic exercises increase skin temperature more?

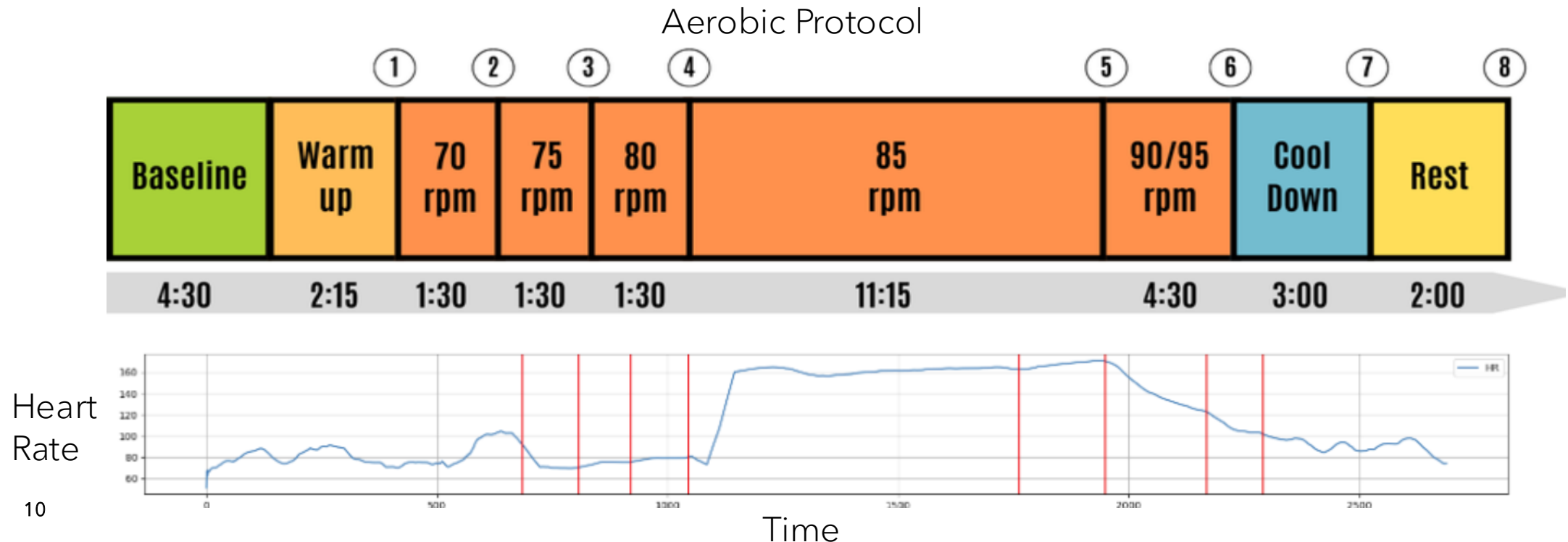
This question interested us because, in humans, sweating can be good for the body. Sweating can be seen to benefit the body by detoxing it from heavy metals, phthalates (plastics), and other chemicals [4]. We want to look at the skin temperature because if the skin temperature is higher a person would sweat more. Answering the question would show support for anaerobic or aerobic exercise being better for detoxing the body if we look at skin temperature. The variables that might be of interest include: Temp.csv from both the anaerobic and aerobic folders.

TASK #3: DATA AND QUESTION ALIGNMENT

Question	Metric	Answers Question(Y/N)
1) On average, how do the heart rates of participants who reportedly do physical activity often, compare to participants who don't do physical activity often; during aerobic exercise?	Heart rate	Yes
2) Does aerobic exercise get better as we age or get worse for individuals who do workout often?	Heart rate	Yes
3) Does arguing for the opposite opinion make a person's heartrate depress or jump when arguing for the opposite opinion?	Heart rate	Maybe
4) For individuals over the average weight in the U.S., how does anaerobic exercise affect heart rate compared to those under average?	Heart rate	Yes
5) Do anaerobic exercise or aerobic exercises increase skin temperature more?	Skin temperature	Yes

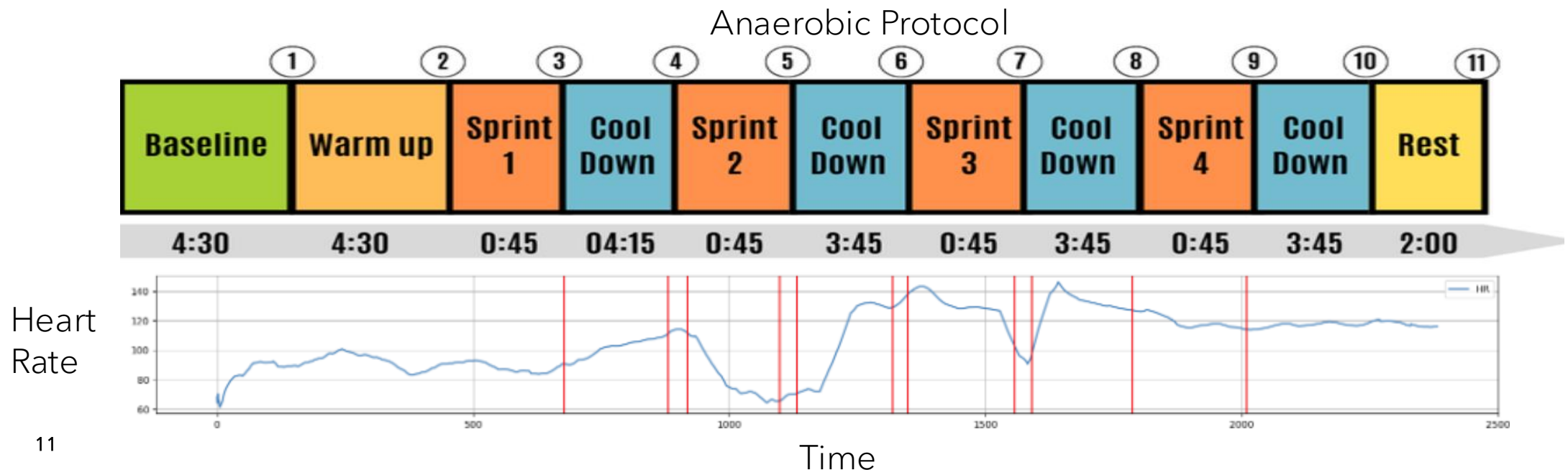
QUESTION #1 DATA & JUSTIFICATION

To compare the heart rates of active and inactive participants during aerobic respiration, we will look at their plots directly across different intervals to make a proper determination.

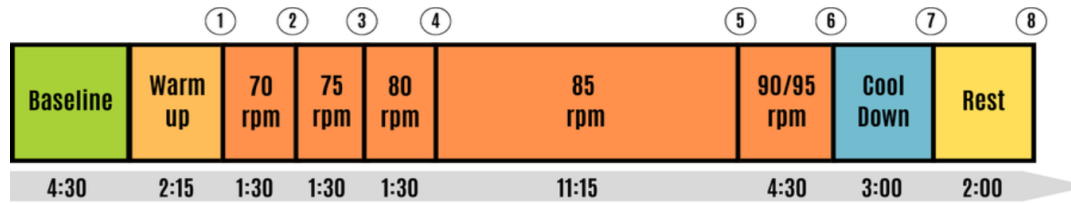


QUESTION #4 DATA & JUSTIFICATION

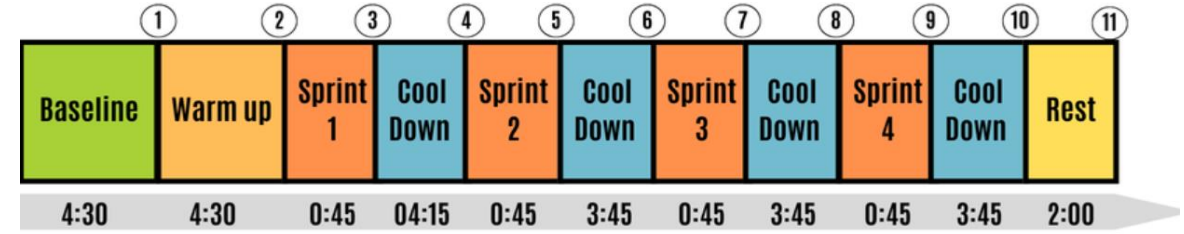
For this comparison, we will utilize the heart rate graphs from different weight classes, as well as create our own graphs using the csv data that represent weight class vs. Average heart rate, to create a better visual of the comparison.



QUESTION #5 DATA



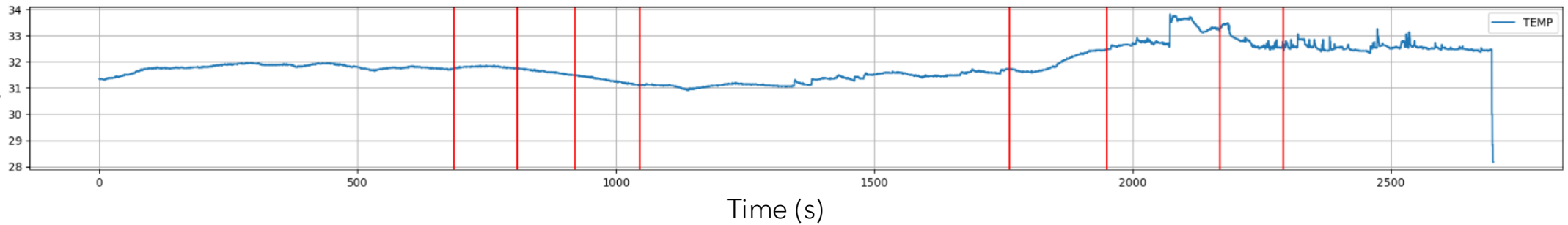
Aerobic Protocol



Anerobic Protocol

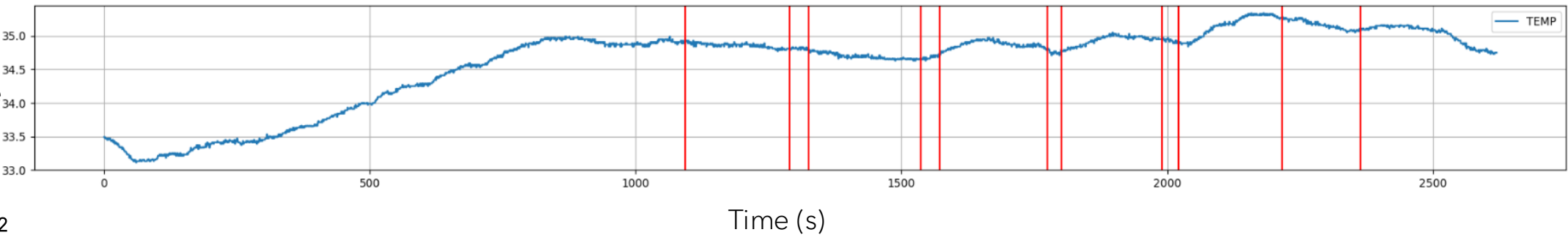
Aerobic (FO1 Subject)

Skin
Temperature
(°C)



Anerobic (FO1 Subject)

Skin
Temperature
(°C)



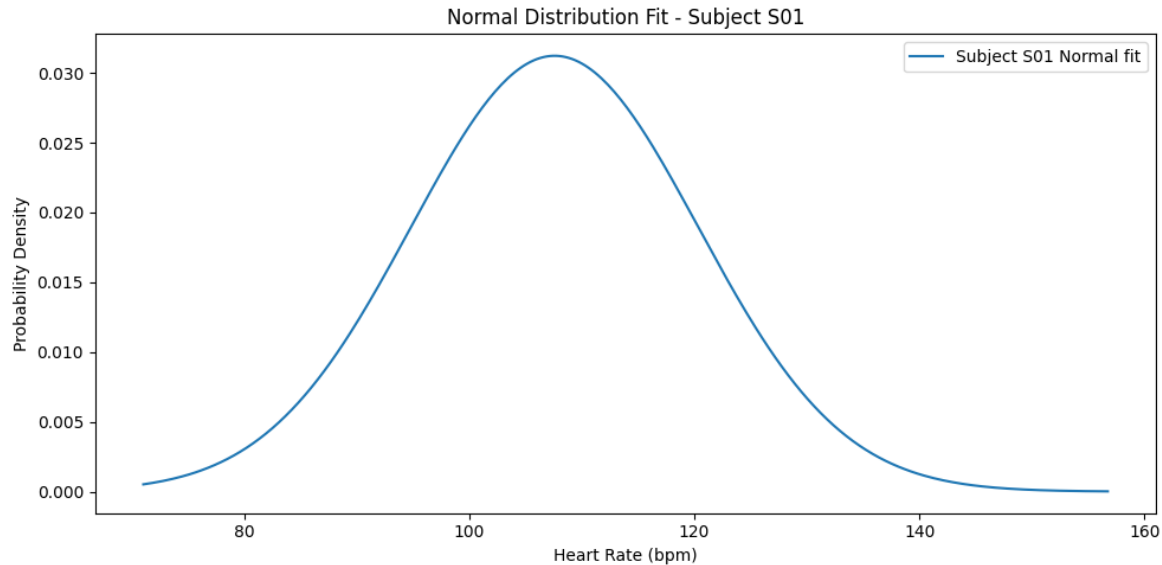
QUESTION #5 JUSTIFICATION

To compare if anerobic or aerobic exercise is better for detoxing the body we will look at skin temperature as this will indicate increased body temperature leading to more sweat. Each test subject has aerobic and anerobic data for skin temperatures as seen in the plots. Notice the two different protocols for anerobic and aerobic exercise. When comparing this data for skin temperature, the data should be analyzed during times when the user is actively exercising.

TASK 4 – METRICS, TOOLS, & PLOTS

	Question #1	Question #4	Question #5
Primary Goal	- To determine if regular exercise habits correlate with a lower/higher average HR during aerobic activity	- To investigate how body weight, relative to the U.S. average, influences HR during anaerobic exercise	- To determine if aerobic or anaerobic elicits higher average skin temp, serving a proxy for sweat production
Data Inputs	- Subject lifestyle surveys - Raw Aerobic HR, time series data	- Subject weight - Anaerobic HR data	- Raw skin temp readings from anaerobic and aerobic activity
Calculations	- Filter subjects into active and inactive groups - Mean HR for each subject - Aggregate mean HR for active vs. Inactive groups - Normal Distribution (μ and σ) - Chi Squared Test	- Categorize subjects into above/below average weight - Extract and average HR from specific patients - Compute mean HR for each group	- Identify subjects that completed both types of exercise - Filter out outliers (readings outside of 23-43 deg C) or (73.4 - Calculate mean skin temp, find global mean
Outputs	- A hypothesis test determining if "fitter" hearts operate healthier during aerobic exercise - Normal distribution curve	- Determination of if higher body mass correlates with higher HR during anaerobic exercise	- Comparison of thermal response - Line graph

TASK 5 – QUESTION 1



----- Goodness of Fit Conclusion -----

Good fits: 0.00 | Not a good fit: 30.00

Conclusion: No subjects had a heart rate that fit a normal distribution.

This implies that heart rate during aerobic exercise is very low at some times, then climbs very high at others

----- Heart Rate Conclusion -----

Subjects who exercise regularly: 98.82 (bpm)

Subjects who do not exercise regularly: 107.65 (bpm)

Conclusion: If you do physical activity regularly, your HEART RATE WILL BE LOWER during Aerobic exercise than if you don't.

Overall Conclusion:

If you do physical activity regularly, your heart rate WILL be LOWER during aerobic exercise than if you don't.

TASK 6 – QUESTION 1

- The original question was answered fully and concluded that heart rate will be lower on average during aerobic exercise for those who are active.
- Though the test group did not follow a normal distribution, the averages correlate with expected values
- We would like to see how a larger dataset would impact the results, likely to see a more normal distribution.

```
----- Normal Distribution -----  
Subject S01: mu = 107.57, std = 12.77  
Subject S03: mu = 122.38, std = 20.46  
Subject S04: mu = 116.14, std = 33.65  
Subject S05: mu = 101.96, std = 16.18  
Subject S06: mu = 88.65, std = 14.90  
Subject S07: mu = 104.65, std = 26.57  
Subject S08: mu = 94.20, std = 16.97  
Subject S09: mu = 85.70, std = 18.02  
Subject S10: mu = 108.27, std = 15.50  
Subject S11: mu = 87.93, std = 7.96  
Subject S13: mu = 112.95, std = 21.30  
Subject S14: mu = 100.77, std = 21.87  
Subject S15: mu = 91.36, std = 5.64  
Subject S16: mu = 99.15, std = 18.83  
Subject S17: mu = 85.59, std = 11.22  
Subject S18: mu = 90.05, std = 17.31  
Subject f03: mu = 85.98, std = 12.34  
Subject f05: mu = 91.01, std = 31.02  
Subject f09: mu = 117.21, std = 37.26  
Subject f10: mu = 84.82, std = 11.07  
Subject S02: mu = 127.35, std = 18.08  
Subject f01: mu = 113.49, std = 37.62  
Subject f02: mu = 86.45, std = 13.22  
Subject f04: mu = 94.53, std = 14.01  
Subject f06: mu = 141.91, std = 32.48  
Subject f07: mu = 130.28, std = 50.06  
Subject f08: mu = 87.31, std = 13.34  
Subject f11: mu = 120.84, std = 17.99  
Subject f12: mu = 86.18, std = 18.84  
Subject f13: mu = 88.16, std = 16.22
```



TASK 5 – QUESTION 4

	Info	Gender	Age	Height (cm)	Weight (kg)	Does physical activity regularly?	Protocol	Stress Inducement	Aerobic Exercise	Anaerobic Exercise
0	S01	m	21	192	84	Yes	V1	Yes	Yes	Yes****
1	S02	m	20	185	95	No	V1	Yes****	Yes	Yes
2	S03	m	20	175	62	Yes	V1	Yes	Yes***	Yes
3	S04	m	21	174	70	Yes	V1	Yes	Yes	Yes
4	S05	m	21	173	72	Yes	V1	Yes	Yes	Yes

Subjects Over Average Weight: ['S01', 'S02', 'S07', 'S14', 'f11', 'f12']

Subjects Under Average Weight: ['S03', 'S04', 'S05', 'S06', 'S08', 'S09', 'S10', 'S11', 'S12', 'S13', 'S15', 'S16', 'S17', 'S18', 'f01', 'f02', 'f03', 'f04', 'f05', 'f06', 'f07', 'f08', 'f09', 'f10', 'f13', 'f14', 'f15', 'f16', 'f17', 'f18']

AVG HR for Subjects Over Average: 90.24236462471964

AVG HR for Subjects Under Average: 97.9634392994392

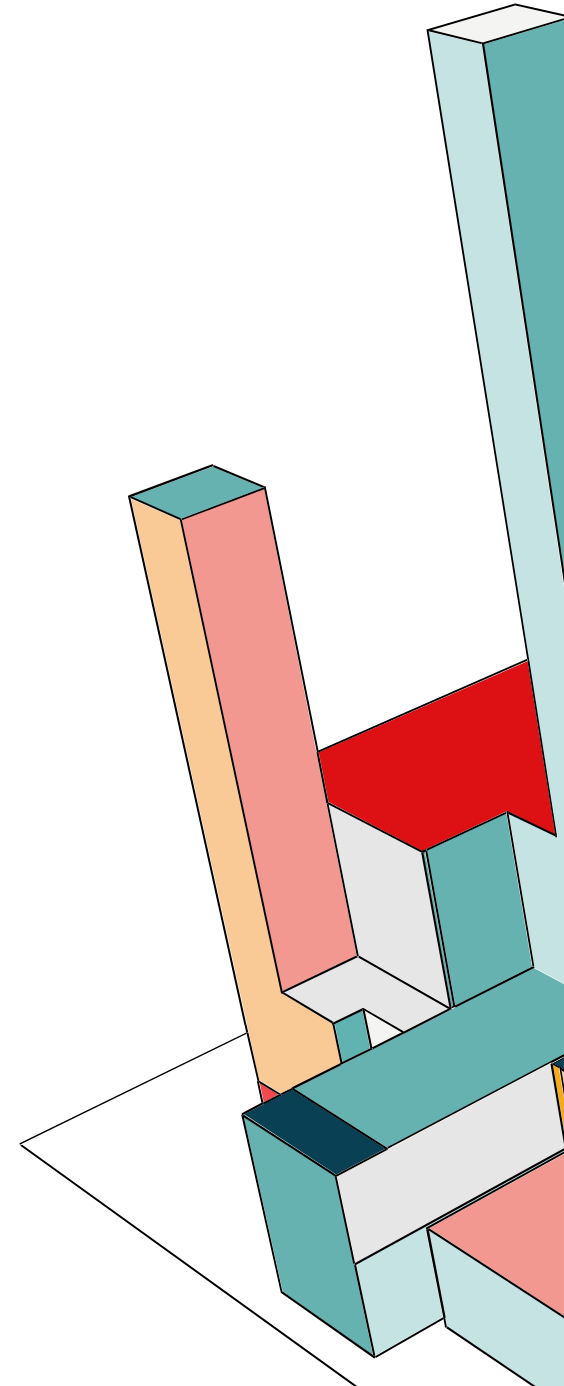
Conclusion: It appears that people who are UNDER the average weight in the U.S. will have a higher heart rate during Anaerobic exercise compared to those OVER the average weight.

Overall Conclusion:

Contrary to popular belief, it appears, based on our data, that people who are under the average weight in the U.S. will have a higher heart rate during anaerobic exercise compared to those above the average weight.

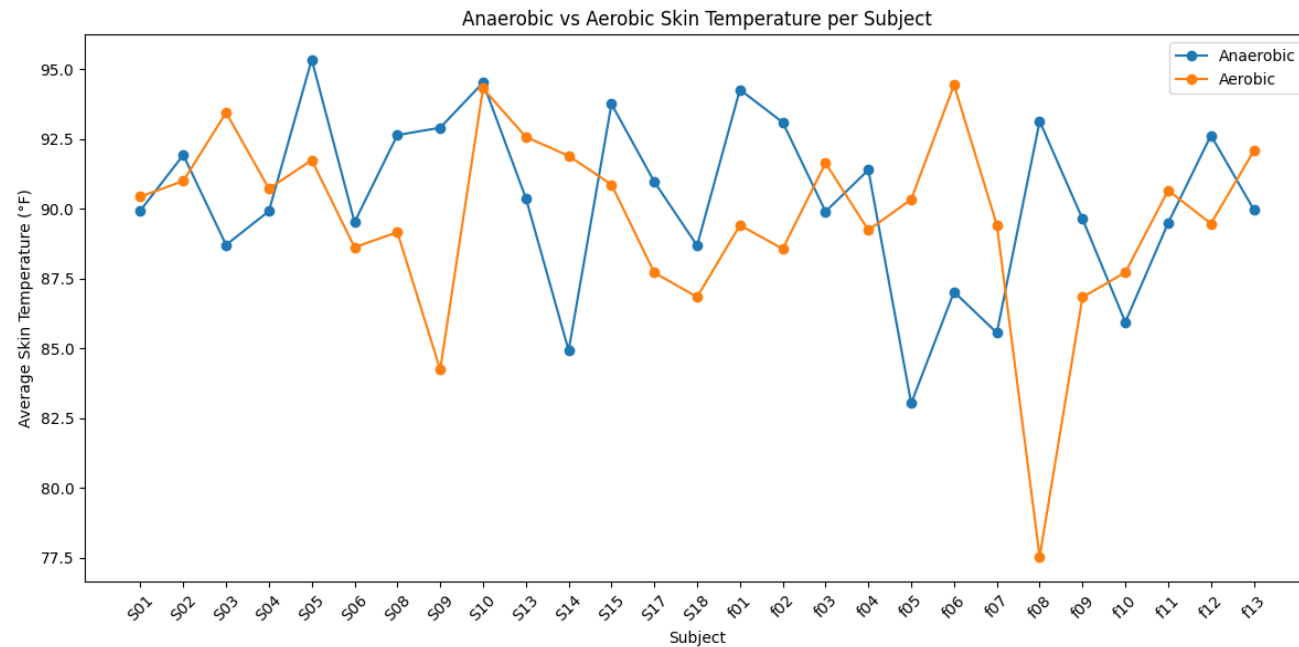
TASK 6 – QUESTION 4

- From the collected data, we see that subjects who are 'underweight' correlate with higher heart rates during anaerobic activity.
- We were surprised by this result, as it would seem that 'underweight' people should have lower heart rates when exercising.
- It is notable that the sample size for the overweight group consisted of 6 subjects, while the underweight group was about 6 times larger.
 - Even though we are analyzing heart rate of the individual, an extremely small sample size of a group could skew the results significantly.



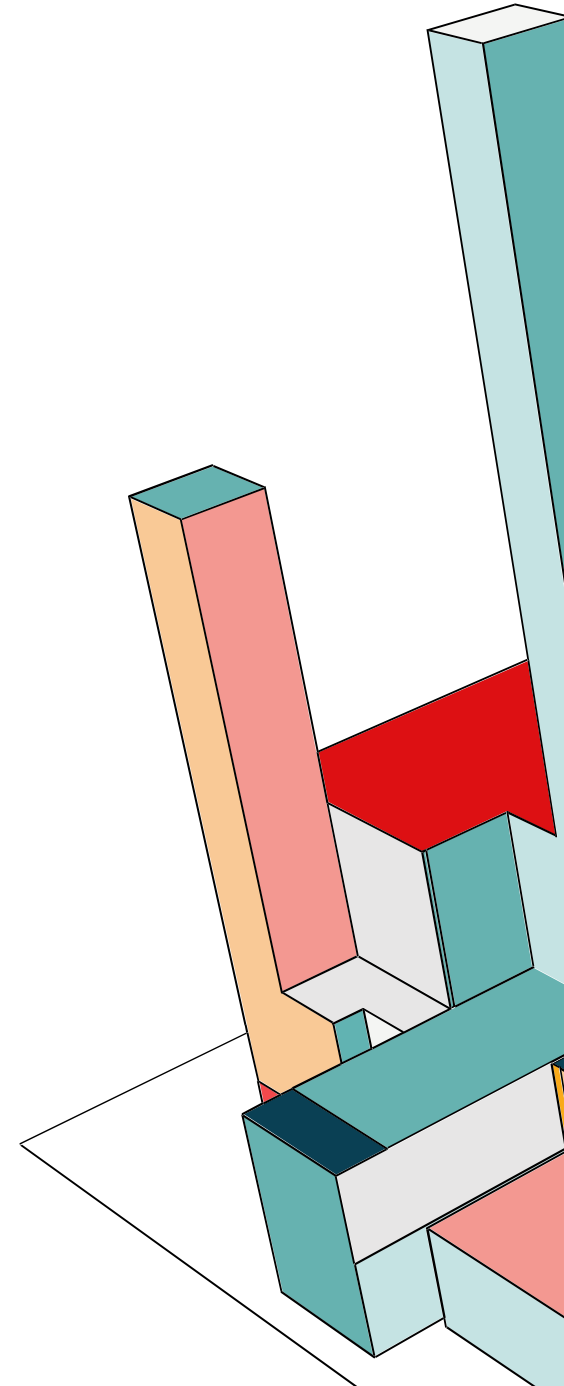
TASK 5 - QUESTION 5

Overall Anaerobic Exercise Skin Temperature: 90.35 °F
Overall Aerobic Exercise Skin Temperature: 89.67 °F
Anaerobic exercise produces higher skin temperature indicating more sweating which could indicate more body detoxification.



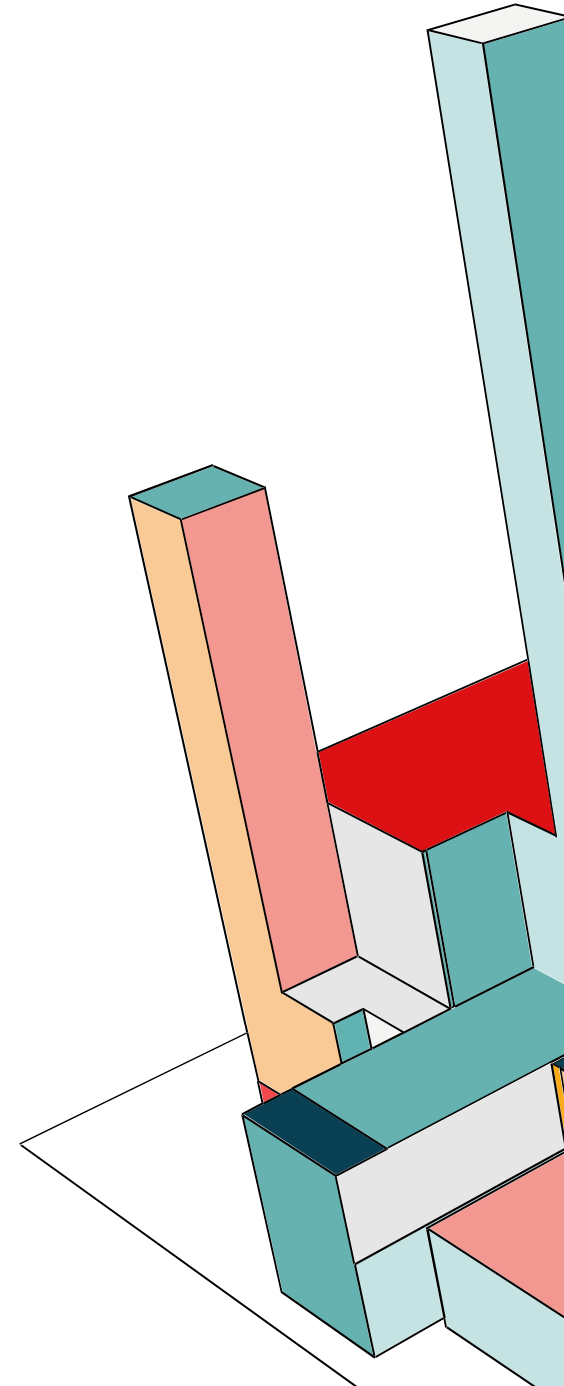
TASK 6 – QUESTION 5

- We conclude from the sample that anaerobic exercise increase skin temperature more than aerobic exercise, which indicates more sweating should occur, which could lead to body detoxification.
- The biggest question that arose from this test has to do with sweating, and not exercise:
 - What does it mean to sweat more?
 - How can we quantize average sweat values?
- In our opinion, a follow-up study should be conducted to yield better results.



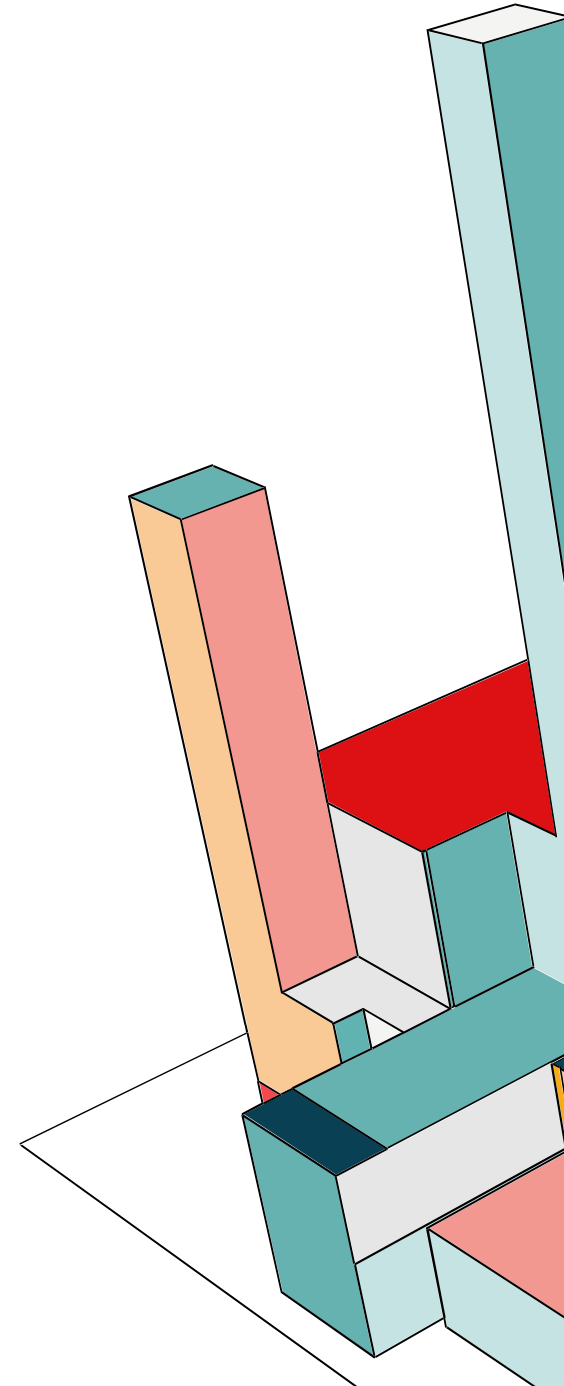
TASK 7 - REFLECTION

- Along the way, we learned:
 - That less values were required to yield the results we seeked
 - Normal distribution and averages are extremely important, but don't tell the whole story
 - Why additional studies are conducted



TASK 7 - REFLECTION

The biggest problems encountered and overcome were extracting the data from the files actually given to us. It wasn't difficult once it was done but to this point most of the files have been easily provided or called in our code, this required some more steps to get that done this time. Another thing would be trying to filter the data and using it to an advantage. For question 4 especially the output of code had useless data, and it didn't make sense because it needed to be filtered. Once issues like that were resolved it made much more sense. The coding world is huge and this course as well as this project opened our eyes to how much programming can be used in as an advantage for numerical data, science, research, and many more.



References

- [1] A. Hongn, F. Bosch, L. Prado, and P. Bonomini, "Wearable Device Dataset from Induced Stress and Structured Exercise Sessions (version 1.0.1)," PhysioNet, 2025. [Online]. Available: <https://doi.org/10.13026/he0v-tf17>
- [2] A. Hongn, F. Bosch, L. E. Prado, J. M. Ferrández, and M. P. Bonomini, "Wearable Physiological Signals under Acute Stress and Exercise Conditions," Scientific Data, vol. 12, no. 1, 2025. [Online]. Available: <https://doi.org/10.1038/s41597-025-04845-9>
- [3] A. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: Components of a new research resource for complex physiologic signals," Circulation, vol. 101, no. 23, pp. e215–e220, 2000. [Online]. Available: <https://doi.org/10.1161/01.CIR.101.23.e215>
- [4] "7 Benefits Of Sweating - Why Breaking A Sweat Is Good For You," Jul. 11, 2022. <https://marathonhandbook.com/7-benefits-of-sweating/>
- [5] C. Meserve, "What is Skin Temperature and Why Should You Monitor It?," WHOOP, Sep. 29, 2021. <https://www.whoop.com/us/en/thelocker/what-is-skin-temperature-and-why-should-you-monitor-it/>

THANK YOU

